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PCI and Reference Generator Documentation

This document describes the theoretical basis and the actual C implementation of the `PCI/PMCI` current controller (`pci.c`) and the coordinated current-reference generator `Reference_Gen` (`reference_gen.c`) used in this project. The control strategy follows:

> C. Zhao, J. Liu, Z. Xie, and F. Zhou, "Coordinate Control of Power/Current for Grid-Connected Inverter Based on PCI Controller under Unbalanced Grid Conditions," *Complexity*, vol. 2019, Article ID 5968984, 14 pages, 2019.

Equation numbers quoted as "Eq. (n) of the paper" refer to that reference. Where the actual implementation extends or departs from the paper — to handle a grid with harmonic distortion, or to obtain an exact filter-free denominator — the difference is called out explicitly, since the code (not the paper) is the final source of truth for this project.

1 Overall System Architecture

The system is a two-level voltage-source inverter (2L-VSI) connected to the grid in *grid-following* mode: the user sets an active power reference P^* and a reactive power reference Q^*, and the control must inject the appropriate current. The full control chain is:

$$\underbrace{(P^*,\, Q^*,\, k)}_{\text{setpoints}} \xrightarrow{\texttt{Reference\_Gen}} \underbrace{i^*_{\alpha\beta}}_{\text{current ref.}} \xrightarrow{\texttt{PCI}} \underbrace{v^*_{\alpha\beta}}_{\text{voltage ref.}} \xrightarrow{\text{SVM/PWM}} \text{VSI} \longrightarrow \text{grid}$$

Two auxiliary blocks feed both stages:

- The **MSOGI-PLL** estimates the grid frequency ω and splits the fundamental voltage into its positive ($\boldsymbol{v}^+$) and negative ($\boldsymbol{v}^-$) sequences in $\alpha\beta$.
- The **Clarke** transform (amplitude-invariant form, 2/3 factor) converts the three-phase *abc* measurements to the stationary $\alpha\beta$ plane.

The division of labour is clean: `Reference_Gen` is the "brain" (it decides *what* current to request given the chosen power strategy), and `PCI` is the "muscle" (it makes sure the actual current tracks that reference with zero steady-state error).

A structural simplification relative to the paper. The paper's control structure (Figs. 2, 3 and 7) runs *two* parallel current-control branches: a fundamental branch with transfer function $G_f(s)$ tracking $i_{\alpha\beta}^{f*}$, and a harmonic branch $G_h(s)$ tracking $i_{\alpha\beta}^{h*} = k\, i_{\alpha\beta}^{f*}$ (Eq. (18)), both compared against the *same* measured current and summed at the modulator (Eq. (21)). In this project, `Reference_Gen` already blends the fundamental and harmonic content into a *single* combined reference $i_{\alpha\beta}^{*}$ (Section 4), so a single PMCI controller with one proportional gain and several resonant terms in parallel is enough: each resonant term locks onto whichever frequency component is actually present in $i_{\alpha\beta}^{*}$, playing the same role as the paper's two separate branches without duplicating the proportional path.

2 Preliminaries: Space Vector and Sequences

2.1 Amplitude-Invariant Clarke Transform

For a three-phase system x_a, x_b, x_c:

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \tag{1}$$

With the $2/3$ factor, *amplitude* is preserved: if $v_a = V\cos(\omega t)$ (balanced system), then $|\boldsymbol{v}_{\alpha\beta}| = V$. The trade-off is that power is *not* invariant, and a $3/2$ factor appears instead:

$$p = \tfrac{3}{2}\left(v_\alpha i_\alpha + v_\beta i_\beta\right), \qquad q = \tfrac{3}{2}\left(v_\beta i_\alpha - v_\alpha i_\beta\right) \tag{2}$$

This is the instantaneous *p*-*q* power definition used by the paper (Eq. (4)). This $3/2$ (and its inverse $2/3$) appears explicitly in `reference_gen.c`.

2.2 The Complex Space Vector

It is convenient to group the two axes into a single complex number:

$$\boldsymbol{v} = v_\alpha + \mathrm{j}\, v_\beta \tag{3}$$

With this notation, a balanced three-phase set of **positive sequence** is a phasor rotating counterclockwise, and one of **negative sequence** rotates clockwise (paper, Eq. (2)–(3)):

$$\boldsymbol{v}^+(t) = V^+ \mathrm{e}^{\mathrm{j}(\omega t + \phi^+)}, \qquad \boldsymbol{v}^-(t) = V^- \mathrm{e}^{-\mathrm{j}(\omega t - \phi^-)} \tag{4}$$

Key geometric property (used by the CBC denominator below): each sequence, on its own, traces a *circle of constant radius* in the $\alpha\beta$ plane, so that

$$|\boldsymbol{v}^+|^2 = (V^+)^2 = \text{const.}, \qquad |\boldsymbol{v}^-|^2 = (V^-)^2 = \text{const.} \tag{5}$$

even though their sum $\boldsymbol{u} = \boldsymbol{v}^+ + \boldsymbol{v}^-$ traces an ellipse whose modulus *does* oscillate in time.

3 The PCI/PMCI Current Controller

3.1 Why a Classical PI Is Not Enough

A PI has transfer function

$$PI(s) = K_p + \frac{K_i}{s} \tag{6}$$

with a pole at $s = 0$: infinite gain *only at DC*. By the internal-model principle, a loop achieves zero steady-state error only for signals whose model (poles) it contains. Here the reference is **sinusoidal at** ω (50 Hz), so a PI in $\alpha\beta$ always leaves a residual amplitude and phase error. The two classical remedies are:

1. **Synchronous dq frame**: rotate the signals with the Park transform so they become DC, then use a PI. It works, but it requires the PLL angle inside the current loop, $\pm\omega L$ decoupling terms, and gets considerably more complex under unbalance (two frames, $+\omega$ and $-\omega$, are needed) and with harmonics.

2. **Stationary $\alpha\beta$ frame**: stay in AC and place the infinite gain *at the signal's own frequency*. This is the family of resonant controllers, to which the PCI belongs.

3.2 The Complex Integrator (CI): Derivation

The idea is to shift the integrator into the frame that rotates with the signal. Let n be the *signed* harmonic order ($+1$ positive-sequence fundamental, -1 negative, -5, $+7$, etc.). Applying the frequency shift $s \to s - \mathrm{j}n\omega$ to the integrator K_i/s:

$$\boxed{CI_n(s) = \frac{K_i}{s - \mathrm{j}\,n\omega}} \tag{7}$$

Its pole sits at $s = \mathrm{j}n\omega$, on the imaginary axis: **infinite gain exactly at frequency $n\omega$ and at the sequence indicated by the sign of n.** This generalises the paper's single-term controller $G_f(s) = K_p + K_i/(s - \mathrm{j}\omega)$ (Eq. (19)) to an arbitrary signed order. The full controller is

$$\boldsymbol{v}^* = K_p\,\boldsymbol{e} \;+\; \sum_i CI_{n_i}(\boldsymbol{e}) \;+\; \boldsymbol{v}_{ff}, \qquad \boldsymbol{e} = \boldsymbol{i}^* - \boldsymbol{i}_{\text{meas}} \tag{8}$$

where $\boldsymbol{v}_{ff}$ is the grid-voltage feedforward. Project naming convention:

- **PCI**: proportional + *one* complex integrator (fundamental only).

- **PMCI**: proportional + *several* complex integrators (fundamental ±1 and selected harmonics), added with `PCI_add_term()`. The paper's own multi-term example, Eq. (20), $G_h(s) = K_{ph} + \frac{K_{i3}}{s-\mathrm{j}3\omega} + \frac{K_{i5}}{s-\mathrm{j}5\omega} + \frac{K_{i7}}{s-\mathrm{j}7\omega}$, lists only unsigned orders $3, 5, 7$. This implementation refines that by assigning to each harmonic the signed order it physically occupies in a three-phase system (the 5th appears in negative sequence, $n = -5$; the 7th in positive sequence, $n = +7$; and so on), so a single resonator per order suffices instead of one per order per sequence.

3.3 From Complex Algebra to Real Scalars: No Complex-Number Library Required

Everything above is written with complex numbers because it is the most compact way to *derive* the controller: it lets a 90° phase shift be written as "multiply by j" instead of juggling two coupled differential equations. But nothing in `pci.c` or `pci.h` is actually a complex type: there is no `complex.h`, no `_Complex double`, and no hand-rolled complex struct with complex-multiply/divide routines. Each term's state is just two plain `doubles`, `s_alpha` and `s_beta` (`COMPLEX_INTEGRATOR` in `pci.h`). The reason this is enough is that every operation performed on the complex state in Eqs. (10)–(11) falls into one of exactly two categories, and both have a trivial real-valued equivalent:

1. **Scaling by a real number** (K_i, T_s, or their product). For $\boldsymbol{s} = s_\alpha + \mathrm{j}s_\beta$ and a real r,

$$r\,\boldsymbol{s} = r\,s_\alpha + \mathrm{j}\,r\,s_\beta,$$

i.e. *no coupling at all* between the two axes: each component is scaled on its own. This is exactly the integration step (`pci.c`, lines 92–93): `s_alpha += Ki*Ts*ec_alpha` and `s_beta += Ki*Ts*ec_beta` are two independent scalar lines, not a single "complex" one.

2. **Multiplying by a unit-modulus phasor** $\mathrm{e}^{\mathrm{j}\varphi} = \cos\varphi + \mathrm{j}\sin\varphi$. This is the *only* place where s_α and s_β actually mix, and expanding the product by hand shows exactly how:

$$(s_\alpha + \mathrm{j}s_\beta)(\cos\varphi + \mathrm{j}\sin\varphi) = (s_\alpha\cos\varphi - s_\beta\sin\varphi) \; + \; \mathrm{j}\,(s_\alpha\sin\varphi + s_\beta\cos\varphi).$$

Reading off the real and imaginary parts term by term gives *precisely* the 2×2 rotation matrix used in the code (`pci.c`, lines 96–100):

$$\begin{bmatrix} s_\alpha \\ s_\beta \end{bmatrix}_{k+1} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} s'_\alpha \\ s'_\beta \end{bmatrix} \tag{9}$$

So "multiplying by a complex phasor" is just standard-library trigonometry: one `cos()`, one `sin()`, and four real multiplies per resonator per step — nothing a complex-arithmetic library would compute any differently, since `cexp`/`csin`/`ccos` would perform this same real sine/cosine evaluation internally anyway. Avoiding the library is not just an optimisation: many embedded and PLECS C-script build targets have limited (or no) support for C99 `_Complex` arithmetic, so writing the controller directly in terms of the (s_α, s_β) pair keeps it portable to exactly those targets. The complex notation of Section 3.2 above is therefore best read as a derivation tool, not a description of the implementation.

3.4 Discretisation: The "Integrate + Rotate" Scheme

Rather than discretising Eq. (7) as a fixed-coefficient filter (a direct, e.g. Forward-Euler, discretisation of the continuous-time cross-coupled realisation shown in the paper's Fig. 4), the code performs two operations per sample on the integrator's complex state $\boldsymbol{s} = s_\alpha + \mathrm{j}s_\beta$ — realised, per Section 3.3, purely with the two real scalars s_α, s_β:

$$\text{1) integrate:} \quad \boldsymbol{s}' = \boldsymbol{s}[k] + K_i\, T_s\, \boldsymbol{e}[k] \tag{10}$$

$$\text{2) rotate:} \quad \boldsymbol{s}[k+1] = \mathrm{e}^{\mathrm{j}\varphi}\, \boldsymbol{s}', \qquad \varphi = n\,\omega\, T_s \tag{11}$$

which, expanded into real coordinates as shown in Section 3.3, are the two independent scaling lines followed by the rotation matrix (9).

Discrete transfer function. Combining (10)–(11):

$$\boldsymbol{s}[k+1] = \mathrm{e}^{\mathrm{j}\varphi}\boldsymbol{s}[k] + K_iT_s\,\mathrm{e}^{\mathrm{j}\varphi}\boldsymbol{e}[k] \implies H(z) = \frac{\boldsymbol{s}(z)}{\boldsymbol{e}(z)} = \frac{K_iT_s\,\mathrm{e}^{\mathrm{j}\varphi}}{z - \mathrm{e}^{\mathrm{j}\varphi}} \tag{12}$$

The discrete pole sits at $z_p = \mathrm{e}^{\mathrm{j}n\omega T_s}$, i.e. **exactly on the unit circle and exactly at frequency** $n\omega$. Two practical consequences follow:

1. **Numerical stability (norm preservation).** The rotation matrix has determinant 1: it neither amplifies nor attenuates the state through rounding. A naive discretisation of the second-order resonant form (e.g. Forward Euler on $s^2 + \omega^2$) can push the poles outside the unit circle (unstable) or detune the peak. Here the pole is on the circle *by construction*, for any T_s and any ω.

2. **Frequency adaptation.** The angle $\varphi = n\omega T_s$ is recomputed *every sample* using the ω delivered by the PLL. If the grid drifts (49.8 Hz, 50.2 Hz, ...), the resonant peak *follows* the grid without re-tuning anything. A fixed-coefficient PR loses its infinite gain as soon as $\omega_{\text{grid}} \neq \omega_{\text{nominal}}$; this robustness is precisely why the PCI is chosen.

3.5 Why Not Just Use a PR?

The PR is a well-established, extensively documented controller, and Section 3.2 already showed that a PR is nothing more than two CIs, at $n = +1$ and $n = -1$, tied to the *same* gain K_i. Given how mature the PR is, two concrete reasons justify building the controller out of individually-addressable CIs instead:

1. **Sequence selectivity is not optional here, it is the point.** A real-coefficient PR transfer function is forced to have a complex-conjugate pole pair: $+\mathrm{j}\omega$ and $-\mathrm{j}\omega$ always come as a matched set, with identical gain. But this project's harmonic bank (Section 3.2) needs $n = -5$ and $n = +7$ *without* their unused conjugate partners $n = +5$ and $n = -7$ — those partners do not correspond to any physical component of a three-phase signal, so a PR placed at 5ω would waste half its poles resonating at a frequency where there is nothing useful to track. A bank of signed CIs puts gain only where the signal actually lives.

2. **Independent, modular tuning.** Each CI carries its own K_i and its own leak/pole-radius (`PCI_set_leak`), and terms are added or removed at runtime with a single `PCI_add_term()` call. Reaching the same selectivity with PRs would need one PR per harmonic *pair* plus a way to zero out the unwanted half of each pair, which is not something a standard PR structure offers — it would require going back to a CI-like decomposition anyway.

None of this makes the PR a bad controller — for a single, always-present, always-symmetric fundamental component (no unbalance, no need to distinguish sequences) a PR is perfectly adequate, simpler to reason about, and just as well documented as claimed. The CI/PMCI structure earns its extra complexity specifically because this project must track an asymmetric, adaptive, multi-frequency reference on an unbalanced, harmonic-polluted grid.

3.6 Loop-Delay Compensation

Digital control introduces a transport delay of ≈ 1.5 samples: 1 sample of computation (measured at k, applied at $k+1$) + 0.5 samples from the PWM zero-order hold. For a signal at frequency $n\omega$, that delay is equivalent to a phase lag

$$\theta_d(n) = n\,\omega\,T_s\,N_d, \qquad N_d \approx 1.5 \tag{13}$$

which **grows linearly with the harmonic order**. The resonator integrates against the "stale" phase and pushes with the wrong phase: at high orders the phase margin of the peak erodes, rejection of the 11th/13th degrades, and the loop can become unstable.

The implemented fix (`pci.c`, lines 80–90) is to *pre-rotate the error* of each resonator by exactly that angle, advancing what the loop is about to delay:

$$\boldsymbol{e}_c = \mathrm{e}^{\mathrm{j}\varphi_c}\,\boldsymbol{e}, \qquad \varphi_c = n\,\omega\,T_s\,N_d \tag{14}$$

3.7 Anti-Windup and Saturation

A resonator has infinite gain: if the actuator saturates (the requested voltage exceeds what the DC bus can synthesise, $|v| > v_{\text{limit}}$), the error persists and the state would grow without bound (*windup*); once out of saturation, the controller would take a long time to "unload" that excess. The code applies two clamps (function `clampd`):

1. **Per integrator** (lines 106–108): each state s_α, s_β is clamped to $\pm v_{\text{limit}}$. Bounds how much each resonator can "charge up".

2. **At the output** (lines 117–118): the total sum $v = K_p e + \sum s_i + ff$ is clamped again to $\pm v_{\text{limit}}$, guaranteeing the reference handed to the modulator is always realisable.

3.8 Summary of the PCI_step Algorithm (pseudocode)

```
e = i_ref - i_meas;                              // vector error
for (each term n_i, Ki_i) {
    ec = rot(n_i*omega*Ts*Nd) * e;               // 0. delay compensation
    s  = s + Ki_i*Ts*ec;                         // 1. integrate
    s  = rot(n_i*omega*Ts) * s;                  // 2. rotate (resonance)
    s  = s * leak;                               // 2b. pole radius <= 1 (damping)
    s  = clamp(s, v_limit);                      // 3. anti-windup
    sum += s;
}
v = clamp(Kp*e + sum + ff, v_limit);             // P + resonators + feedforward
```

Listing 1: Structure of PCI_step (simplified)

4 The Coordinated Reference Generator (Reference_Gen)

4.1 The Problem: From Power to Current on a Non-Ideal Grid

We want to solve for the current $\boldsymbol{i}^*$ that delivers exactly (P^*, Q^*). From Eq. (2), treating it as a 2×2 linear system in (i_α, i_β) and inverting it (paper, Eq. (5)):

$$\boxed{i_\alpha^* = \frac{2}{3}\,\frac{P^* v_\alpha + Q^* v_\beta}{v_\alpha^2 + v_\beta^2}, \qquad i_\beta^* = \frac{2}{3}\,\frac{P^* v_\beta - Q^* v_\alpha}{v_\alpha^2 + v_\beta^2}} \tag{15}$$

This is the solution given by **instantaneous (p-q, Akagi) power theory** (paper, Eq. (6)). Note the structure: a *numerator* combining P, Q with the voltage, and a *denominator* $|\boldsymbol{v}|^2$. **The whole design problem reduces to deciding which voltage to use at each spot**, because on a non-ideal grid an unavoidable conflict appears.

4.2 The Conflict: Why $|\boldsymbol{u}|^2$ Oscillates at 2ω

Let the unbalanced fundamental voltage be $\boldsymbol{u} = \boldsymbol{v}^+ + \boldsymbol{v}^-$ per Eq. (4). Its squared modulus:

$$\begin{aligned} |\boldsymbol{u}|^2 = \boldsymbol{u}\,\boldsymbol{u}^* &= \left(V^+ e^{j\omega t} + V^- e^{-j\omega t}\right)\left(V^+ e^{-j\omega t} + V^- e^{j\omega t}\right) \\ &= \underbrace{(V^+)^2 + (V^-)^2}_{\text{constant}} + \underbrace{2\,V^+ V^- \cos(2\omega t + \phi)}_{\text{ripple at } 2\omega} \end{aligned} \tag{16}$$

(phases absorbed into ϕ; this is the paper's Eq. (7) up to the phase convention chosen for ϕ). The cross term between sequences produces a **ripple at twice the grid frequency**. This is where the two pure strategies come from:

- If Eq. (15) uses the *instantaneous* denominator $|\boldsymbol{u}|^2$ (with its ripple), the division exactly compensates the oscillation: **power comes out constant**, but the current inherits the denominator's ripple at 2ω (it becomes **distorted**, with a dominant 3rd harmonic). This is the paper's *instantaneous power control* (Eq. (6)).
- If a *constant* denominator is used instead, the current keeps the shape of the numerator (**sinusoidal, low THD**), but since the voltage does oscillate and the current does not, the **delivered power ripples at** 2ω. This is the paper's *current balance control* (Eq. (8)).

Strategy	Denominator	Current	Power
IPC ($k = 1$)	$\lvert\boldsymbol{u}\rvert^2$, instantaneous	distorted	constant
CBC ($k = 0$)	constant (see Section 4.3)	sinusoidal, low THD	ripples at 2ω

You cannot have both at once: it is a **structural trade-off** of any inverter on an unbalanced grid, not a limitation of the control.

4.3 The Elegant Detail: An Exact, Filter-Free CBC Denominator

The naive way to obtain a constant denominator would be to low-pass filter the instantaneous $|\boldsymbol{u}|^2$ to keep only its mean, but that introduces *delay* and pollutes transients. The code avoids the filter entirely using Eq. (5): since the MSOGI-PLL already delivers the separated sequences, and each sequence has *constant modulus on its own*,

$$\mathrm{den}_{dc} = |\boldsymbol{v}^+|^2 + |\boldsymbol{v}^-|^2 = (V^+)^2 + (V^-)^2 \tag{17}$$

is **exactly** the DC component of Eq. (16), computed algebraically, instantaneously and without lag. The 2ω ripple comes from the cross term $2V^+V^-\cos(2\omega t + \phi)$, which is simply *not included*. Likewise, the fundamental voltage is reconstructed as $\boldsymbol{u} = \boldsymbol{v}^+ + \boldsymbol{v}^-$, also exact and delay-free (the DSOGI bank gives a clean fundamental).

Deviation from the paper's original current-balance strategy. The paper's current-balance control (Eq. (8)) uses *only the positive sequence* $\boldsymbol{v}^+$, in *both* numerator and denominator:

$$\boldsymbol{i}^*_{\text{CBC, paper}} = \tfrac{2}{3}\,\frac{P^*\boldsymbol{v}^+}{(V^+)^2} \text{ (plus the analogous } Q^* \text{ term)}$$

which forces the current to be *purely positive-sequence*: this is literally why the paper calls it "current balance" and why Table 2 of the paper credits it with a perfectly balanced current. This project's $k = 0$ current, by contrast, uses the *full* reconstructed fundamental $\boldsymbol{u} = \boldsymbol{v}^+ + \boldsymbol{v}^-$ in the numerator, divided by Eq. (17):

$$\boldsymbol{i}^*_{\text{CBC}} = \tfrac{2}{3}\,\frac{P^*\boldsymbol{u}}{(V^+)^2 + (V^-)^2} \text{ (plus the analogous } Q^* \text{ term)} \tag{18}$$

Since $\boldsymbol{u}$ is itself purely at frequency ω (a sum of a $+\omega$ and a $-\omega$ phasor, with no higher harmonics), this current is still **harmonic-free**, matching the paper's "sinusoidal, low THD" property. It is *not*, however, purely positive-sequence: it retains a small negative-sequence component proportional to the grid's own unbalance ($\propto V^-$). In short, this implementation's $k = 0$ strategy guarantees an exact, filter-free, harmonic-free reference, but — unlike the paper's original CBC — it does not force the injected current to be perfectly balanced. This is a deliberate simplification: it needs only the DSOGI sequence magnitudes (already available from the PLL) and no separate positive-sequence-only current path.

4.4 The Coordinated Blend: The k Coefficient

Instead of picking one extreme, the two references are linearly interpolated with $k \in [0, 1]$ (paper, Eqs. (17)–(18), Section 4):

$$\boxed{\boldsymbol{i}^* = \boldsymbol{i}^*_{\mathrm{CBC}} + k\,(\boldsymbol{i}^*_{\mathrm{IPC}} - \boldsymbol{i}^*_{\mathrm{CBC}})} \tag{19}$$

- $k = 0$: pure CBC — clean current, all the ripple goes into the power.
- $k = 1$: pure IPC — flat power, all the ripple goes into the current.
- $0 < k < 1$: the ripple is *shared* continuously. This lets a designer, for instance, meet a current-THD limit and a DC-bus power/voltage-ripple limit simultaneously, by choosing the smallest k that satisfies both.

Because both strategies share the same numerator (`num_alpha`, `num_beta` in the code), the blend is mathematically equivalent to interpolating the *inverse of the denominator*, and the mean value of power is preserved at P^*, Q^* for every k.

Relation to the paper's realisation. The paper realises this same weighted trade-off with *two parallel resonant-filter branches* ($G_f(s)$, $G_h(s)$, Fig. 3) acting on a single instantaneous-power-control current, one branch keeping only its fundamental content and the other only its harmonic content (Eqs. (9)–(16) of the paper prove that the fundamental component of the instantaneous-power-control current equals exactly the paper's own current-balance reference). This project instead evaluates the CBC and IPC references directly in $\alpha\beta$ and blends them algebraically *outside* the current loop, in `Reference_Gen`, so no extra filter branch is needed inside the control loop. For the paper's original (positive-sequence-only) CBC definition the two realisations coincide exactly; with the generalised, filter-free denominator of Section 4.3 they coincide only approximately, to the same order as the grid's own unbalance ratio V^-/V^+.

4.5 Fundamental IPC vs. Wideband IPC

There is a second, independent choice: constant with respect to *which* voltage? The paper's own grid model (Eq. (1)) contains only a positive and a negative fundamental component, with no explicit harmonic voltage content, so this distinction does not arise there. It is introduced in this project to cooperate with the MSOGI harmonic bank on a grid that genuinely has voltage harmonics:

- **Fundamental IPC** (`ipc_wideband = 0`): uses only the *fundamental* voltage $\boldsymbol{u} = \boldsymbol{v}^+ + \boldsymbol{v}^-$ in numerator and denominator (this is the direct implementation of the paper's Eq. (6)/(9), using the DSOGI-reconstructed fundamental instead of the raw measurement). Power is constant *against the fundamental*; but if the grid has harmonics (5th, 7th, ...), they are not in $\boldsymbol{u}$, and the *real* power ($\boldsymbol{v}_{\text{real}} \cdot \boldsymbol{i}$) still ripples. "Partial" constancy, moderately distorted current.

- **Wideband IPC** (`ipc_wideband = 1`): numerator *and* denominator use the full instantaneous voltage $\boldsymbol{v}_{wb}$ (the raw Clarke output, with every harmonic). This is the paper's Eq. (6) formula applied directly to the real, harmonic-polluted measurement: p and q come out constant against the **real** voltage. The price: the current must "undo" every voltage harmonic, comes out noticeably more distorted, and the inverter (PCI bandwidth + switching frequency) must be able to synthesise it.

4.6 Summary of the `REFERENCE_GEN_step` Algorithm (pseudocode)

```
u = v_pos + v_neg;                              // exact fundamental, no lag
num   = { P*u_a + Q*u_b,  P*u_b - Q*u_a };      // shared p-q numerator
den_i = u_a^2 + u_b^2;                          // IPC: instantaneous (ripples at 2w)
den_dc= |v_pos|^2 + |v_neg|^2;                  // CBC: exact DC (Eq. circulos)
den_* = max(den_*, 1.0);                        // start-up floor

i_dl = (2/3) * num / den_dc;                    // CBC  (k=0)
if (ipc_wideband)                               // IPC  (k=1)
    i_gl = (2/3) * num_wb / |v_wb|^2;           //   against the real voltage
else i_gl = (2/3) * num    / den_i;             //   against the fundamental

i_ref = i_dl + k*(i_gl - i_dl);                 // coordinated blend
p_check = 1.5*(u_a*i_a + u_b*i_b);              // open-loop check
q_check = 1.5*(u_b*i_a - u_a*i_b);
```

Listing 2: Structure of REFERENCE_GEN_step (simplified)

The block is **purely algebraic (stateless)**: unlike the PCI, it keeps nothing between steps, so it can run entirely in PLECS's *Output* cell without the copy–step–commit pattern.

5 Equation ↔ Code Correspondence

5.1 pci.c

Lines	Code	Equation / concept
71–72	`e = i_ref - i_meas`	vector error $\boldsymbol{e}$
80–90	pre-rotation of $\boldsymbol{e}$	delay compensation (14)
92–93	`s += Ki*Ts*ec`	integration (10)
96–100	cos / sin matrix	rotation (11), pole at $e^{jn\omega T_s}$
102–104	`s = leak*rot(s)`	pole-radius damping (<1: light leak)
106–108	`clampd(s)`	per-integrator anti-windup
114–115	`Kp*e + sum + ff`	control law (8)
117–118	`clampd(v)`	output saturation

5.2 reference_gen.c

Lines	Code	Equation / concept
29–30	`u = v_alpha_pos + v_alpha_neg` (and β)	exact fundamental
32–33	`num_alpha, num_beta`	p-q numerator (15)
35	`den_inst`	$\lvert\boldsymbol{u}\rvert^2$ with ripple (16)
36–37	`den_dc`	exact DC (17)
39–40	floor 1.0	start-up protection
43–44	`i_alpha_dl, i_beta_dl`	CBC ($k = 0$)
52–64	`i_alpha_gl, i_beta_gl`	fundamental / wideband IPC
67–68	`i_alpha_ref, i_beta_ref`	blend (19)
73–74	`p_check, q_check`	verification (2)

6 Final Synthesis

- **Reference_Gen** translates (P^*, Q^*) into current by inverting p-q theory (15), and exposes with a single knob k the structural trade-off of unbalanced grids: clean current (CBC) ↔ flat power (IPC). Its central refinement is the *exact* DC denominator (17), obtained from the sequences with no filter and no lag — at the cost, relative to the paper's original current-balance strategy, of allowing a small residual negative-sequence current proportional to the grid's own unbalance (Section 4.3).

- **PCI/PMCI** makes the actual current track that reference with zero steady-state error, by placing infinite-gain poles at $jn\omega$ (7) via the "integrate + rotate" scheme (10)–(11), which is numerically stable (pole on the unit circle by construction) and self-tunes with the PLL's ω. Delay compensation (14) preserves rejection of the high harmonics, and per-integrator anti-windup keeps the control recoverable after saturation. A single PMCI loop plays the role of the paper's two parallel branches (G_f, G_h) because `Reference_Gen` already hands it one combined reference.

- Together they form the chain $(P^*, Q^*, k) \rightarrow i^* \rightarrow v^*$: power strategy on top, exact tracking underneath — both grounded in Zhao et al., *Complexity* 2019, and extended where the project's grid model (voltage harmonics, an exact filter-free CBC denominator) goes beyond what the paper considers.

References

[1] Zhao, C., Liu, J., Xie, Z., & Zhou, F. (2019). Coordinate Control of Power/Current for Grid-Connected Inverter Based on PCI Controller under Unbalanced Grid Conditions. *Complexity*, *2019*, Article ID 5968984, 14 pages. `https://doi.org/10.1155/2019/5968984`